Group 7

SINGAPORE CRIMINAL LAW ADVISORY SYSTEM

An advisory and decision-support tool that helps users and their counsel quickly understand a criminal matter.

Problem

Complexity

Singapore criminal law outcomes depend heavily on specific statutes, court procedure, sentencing principles, and prior case precedent, **making initial case assessment time-consuming** even for lawyers.

Unreliable AI

General-purpose language models **cannot be trusted** for this task, as they may hallucinate authorities, confuse jurisdictions, or produce answers not grounded in Singapore law.

Gap

There is **no domain-specific tool** that provides a fast, structured, evidence-based first-pass assessment of a Singapore criminal matter.

Problem Statement

How might we design a **domain-specific system** that can provide structured and evidence-based **initial legal guidance** for **Singapore criminal law cases?**

System Purpose & Scope

Limited to Singapore Criminal Law

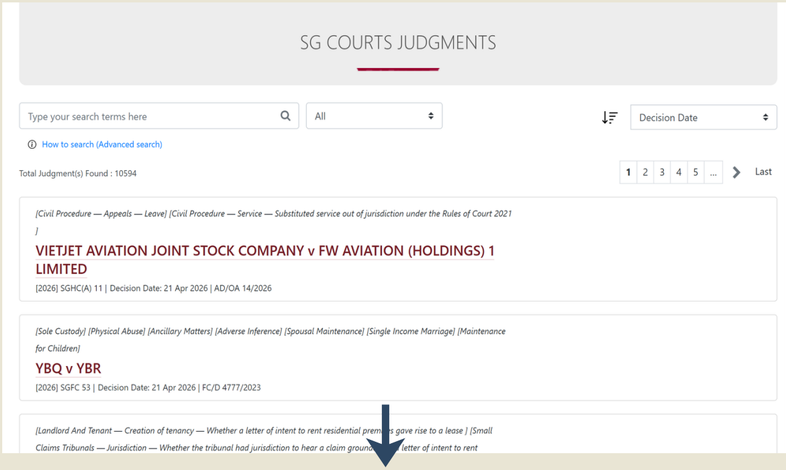
- Available dataset is sourced from **Singapore eLitigation cases**
- Narrowing the **focus to criminal law** helps reduce complexity and makes the problem more manageable

Designed as a decision support tool

- Rather than having a fully autonomous legal advisor, legal reasoning in criminal law requires **careful human judgment,** contextual interpretation, and accountability
- Speed up understanding, while leaving final interpretation, strategy, and responsibility to legal professionals

Raw Dataset

eLitigation (SUPCT)



collect_links():
collect URLs that contains civial cases

↓

scrape_case():
title, citation, catchwords per case

↓

is_criminal_case():
fliter & only let criminal cases pass

→

download_pdf():
save cases as pdf

↑

classify_catchword():
taxonomy classification

↑

dataset.csv:
dataset of 876 cases & 11 features

- **Official court source:**
  - Judgments and case records from Singapore courts
- **Domain-specific content:**
  - Singapore law across various domains
- **Text-heavy and unstructured nature:**
  - Raw data is primarily long-form text, requiring preprocessing, chunking, and NLP techniques to extract usable information

Taxonomy Classification

Structured Taxonomy

- Defines legal categories:
  - **Area → Topic → Subtopic → Statute**
- Each entry maps a known legal catchword pattern to structured labels

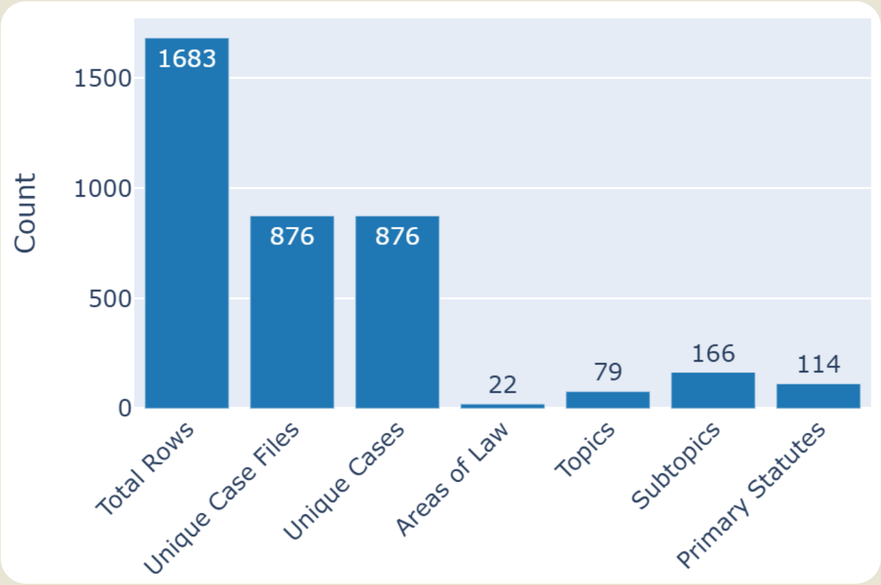
Longest-prefix Matching

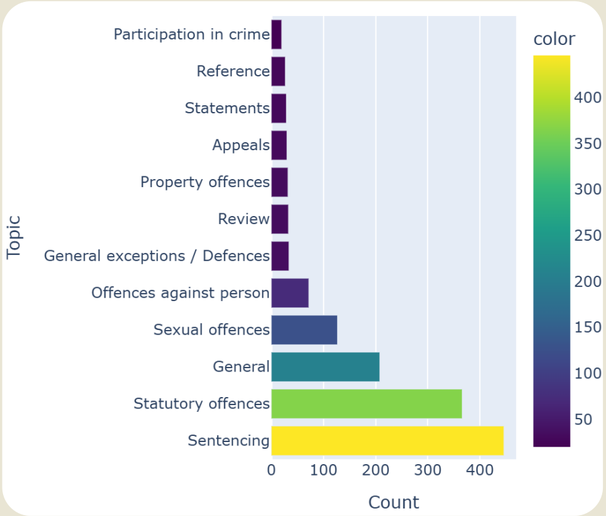
- Normalize input (case-insensitive, standardize dashes like —, –, -)
- Match **the longest prefix of the catchword against taxonomy entries**
  - Ensures more specific classifications override generic ones

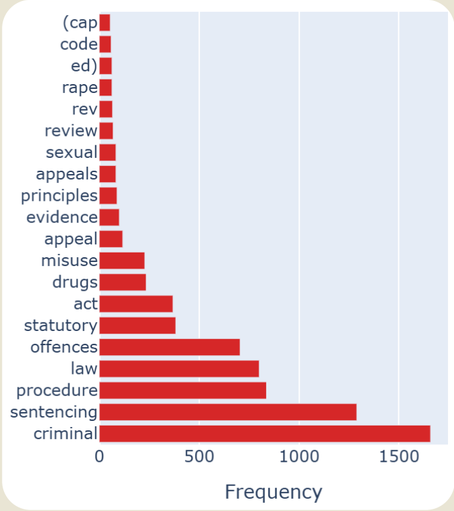
Fallback & Criminal Detection

- If no match:
  - split_catchword(raw) breaks the string into parts using delimiters
  - Assigns best-effort structured labels
- Uses keyword rules
- Filters or flags cases as criminal before downloading or processing

Exploratory Data Analysis







876

unique cases

1,683 rows × 11 cols

98.0%

taxonomy match

up from 78.5% pre-fix

83K

indexed chunks

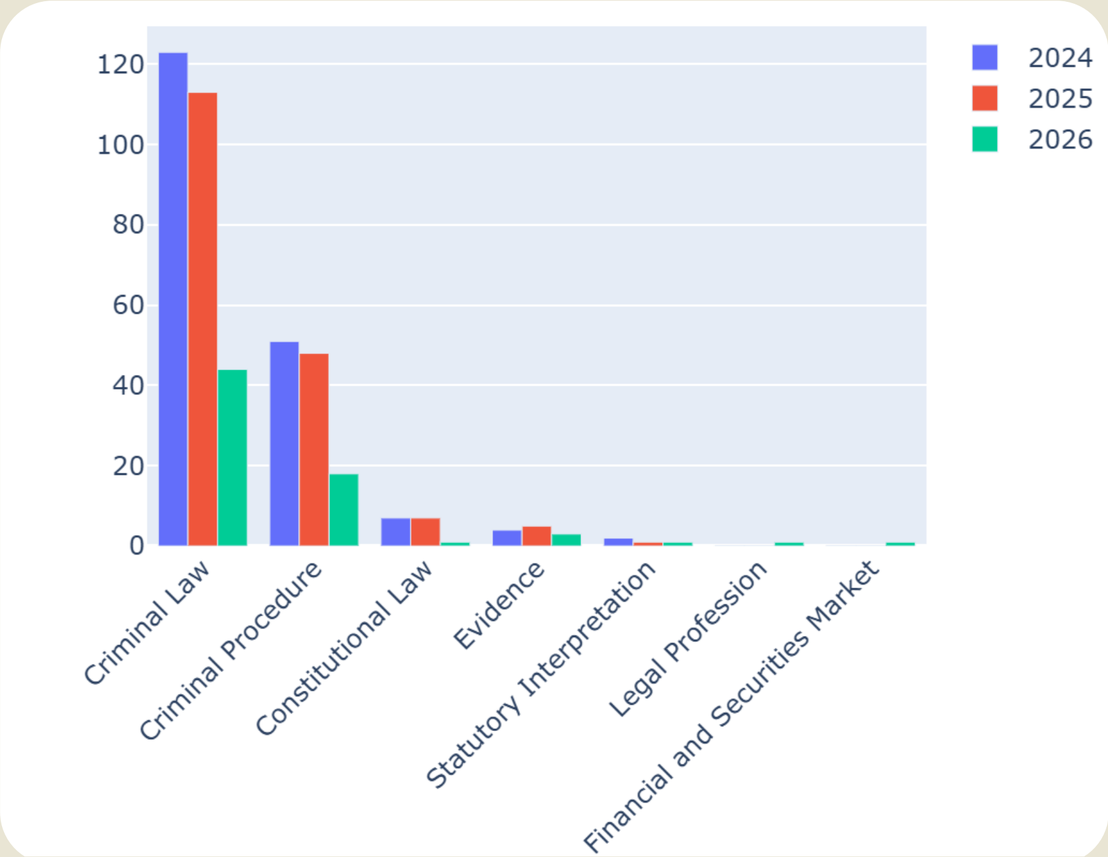
across 8 domains

2015–26

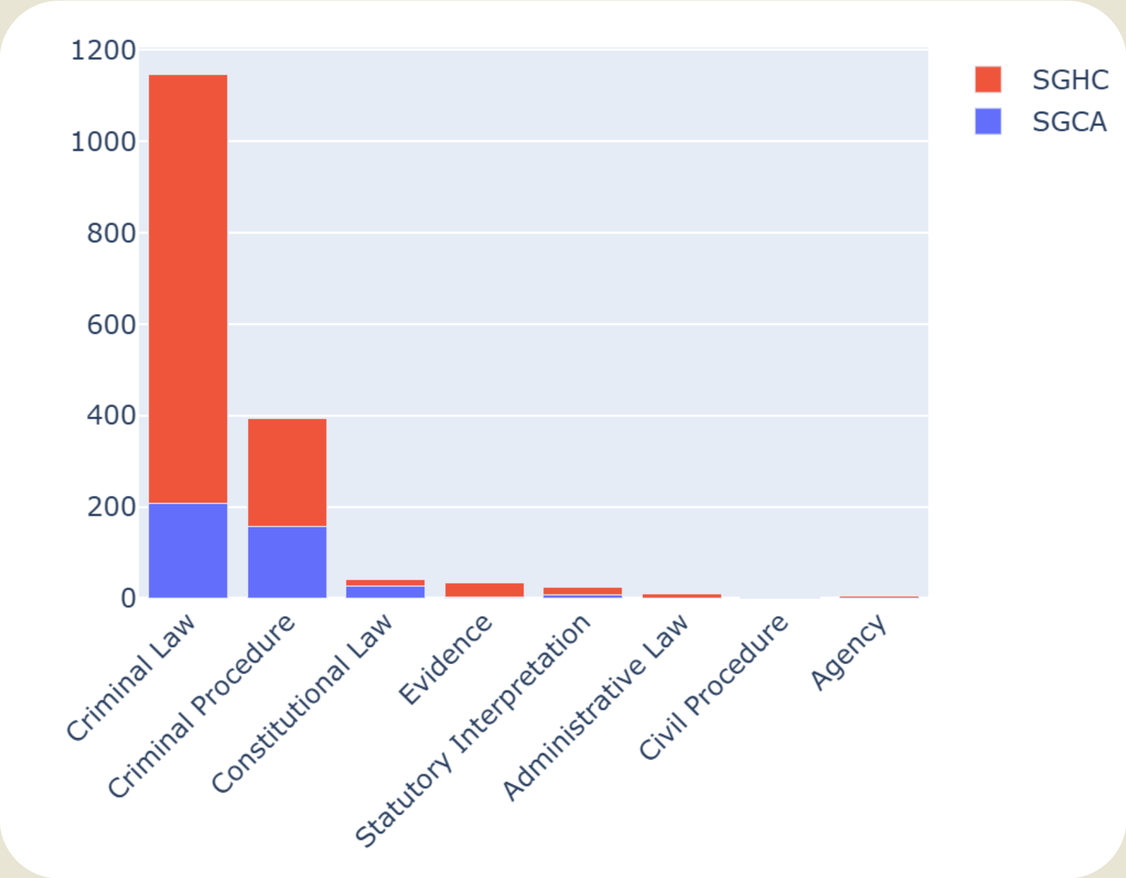
date coverage

SGHC, SGCA, SGHCF

Exploratory Data Analysis

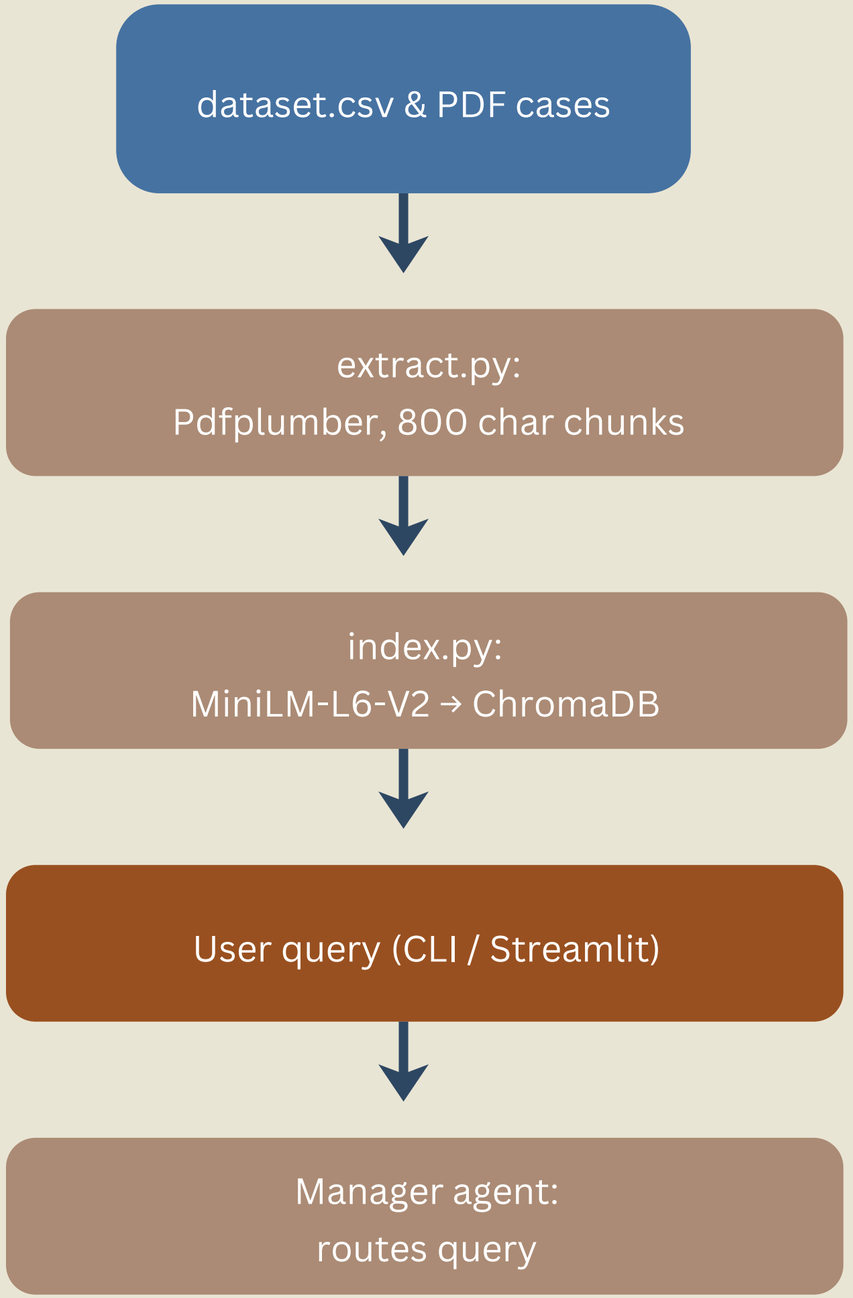


Top 10 Areas of Law in Recent Years

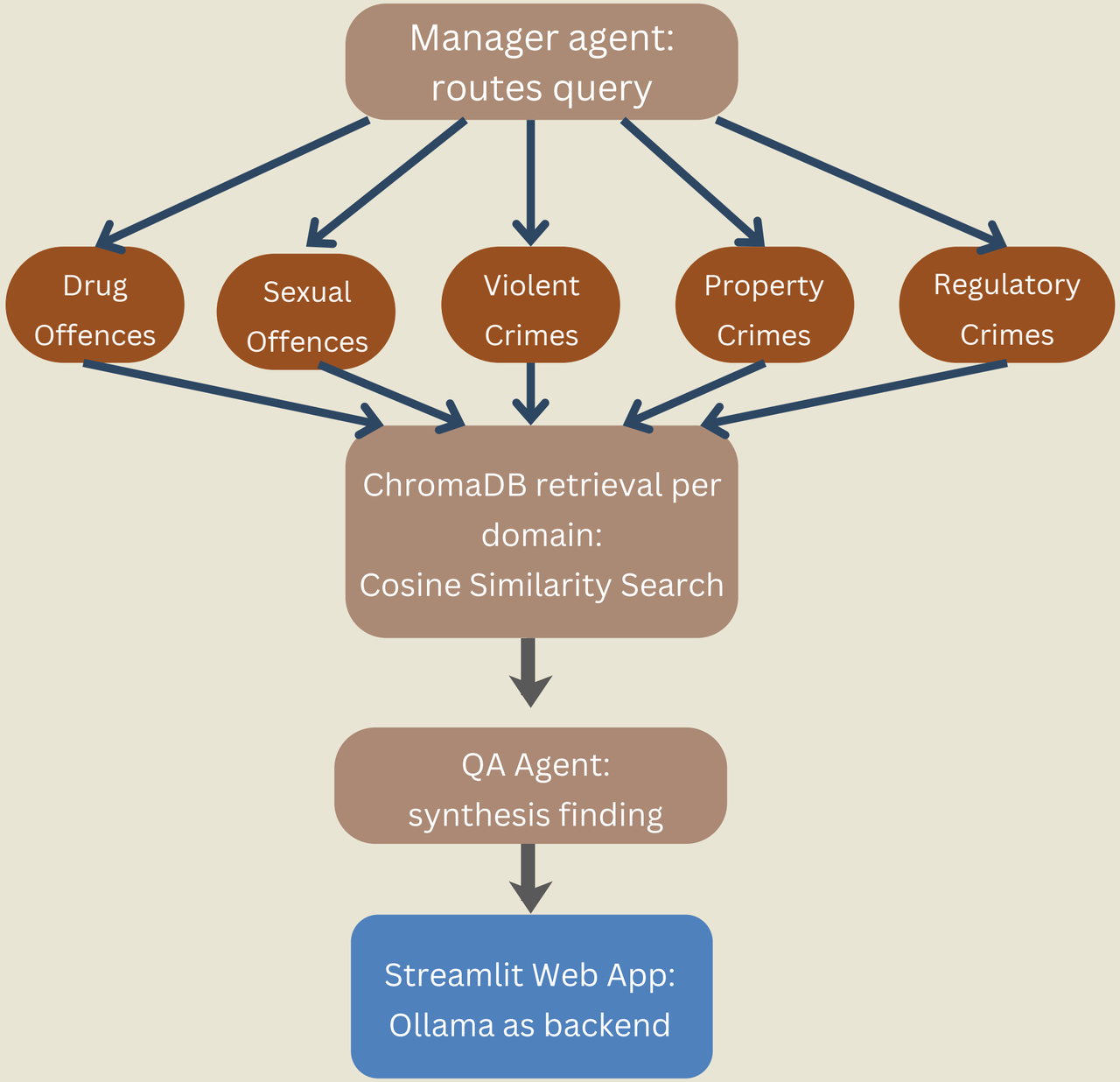


Top 8 Areas of Law by Court Level

Agentic RAG Pipeline



Expert Agents



Seven specialist expert agents, one per domain

Domain specialists (routed by Manager):

- Drug Offences
- Sexual Offences
- Violent Crimes
- Property & Financial
- Regulatory

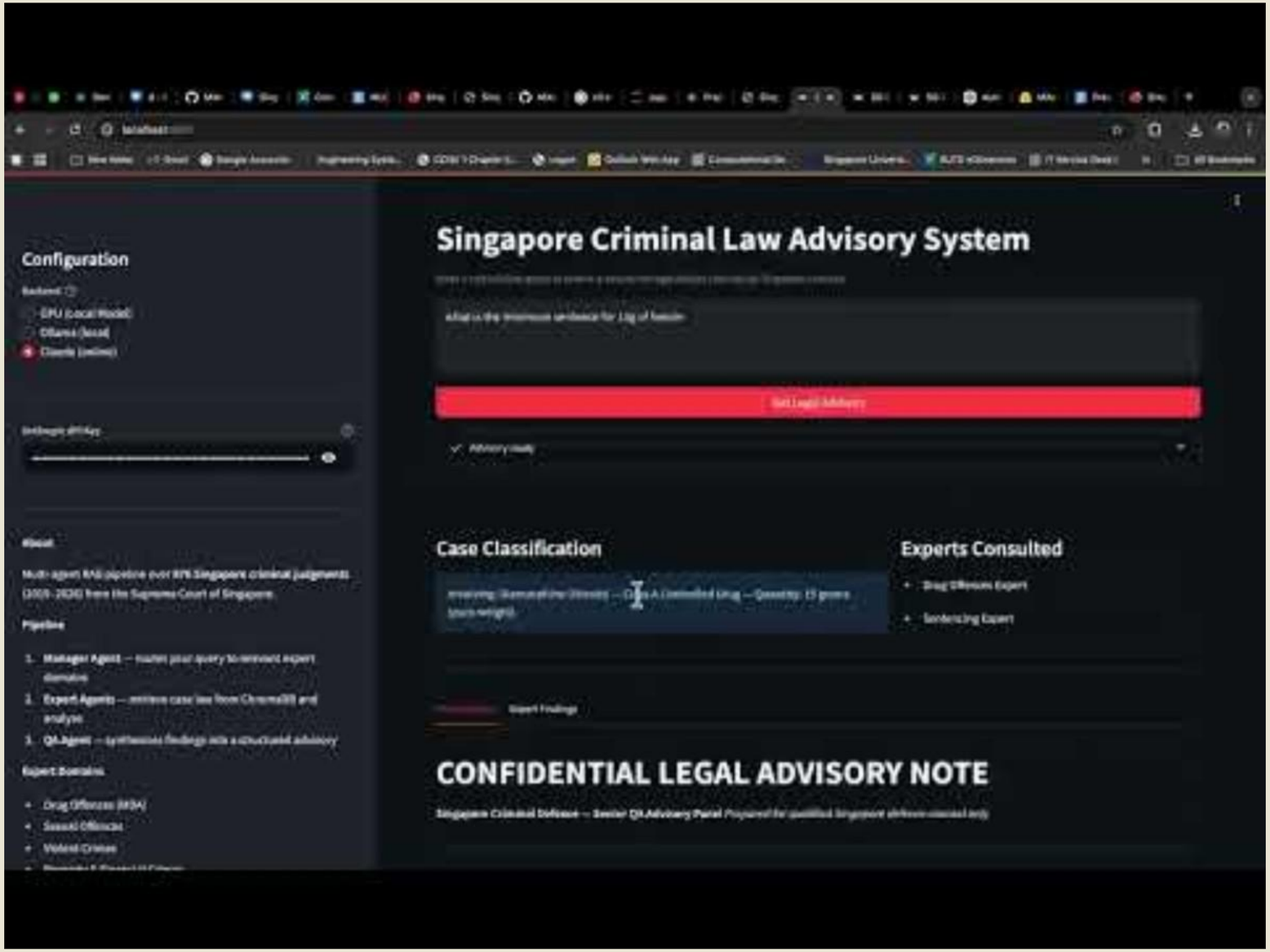
Always consulted for any criminal query:

- Sentencing
- Criminal Procedure

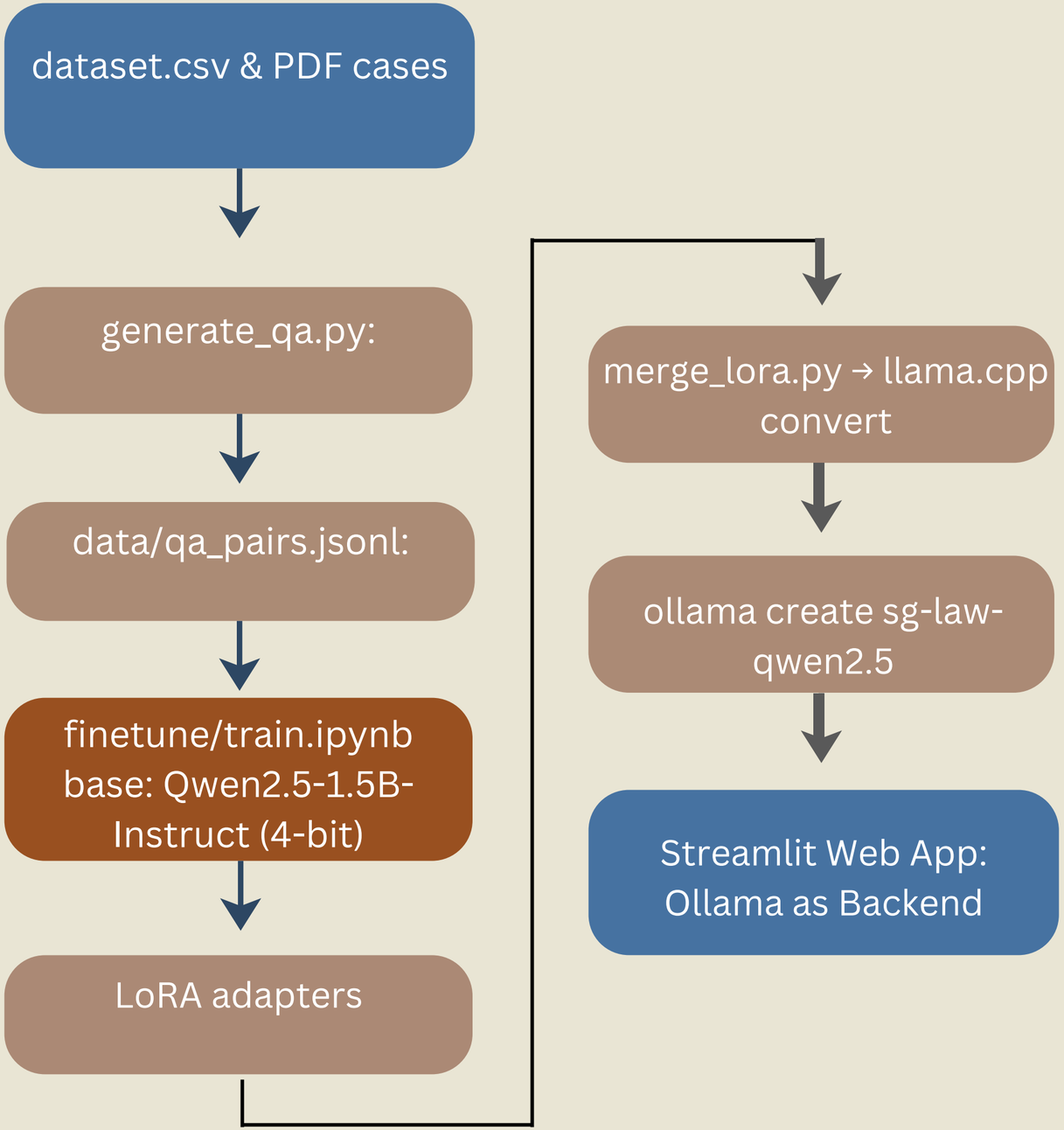
Each expert agent:

- Retrieves the top-5 most relevant chunks from its ChromaDB collection
- Calls the LLM backend with a domain-specific system prompt
- Returns structured findings + Singapore neutral citations

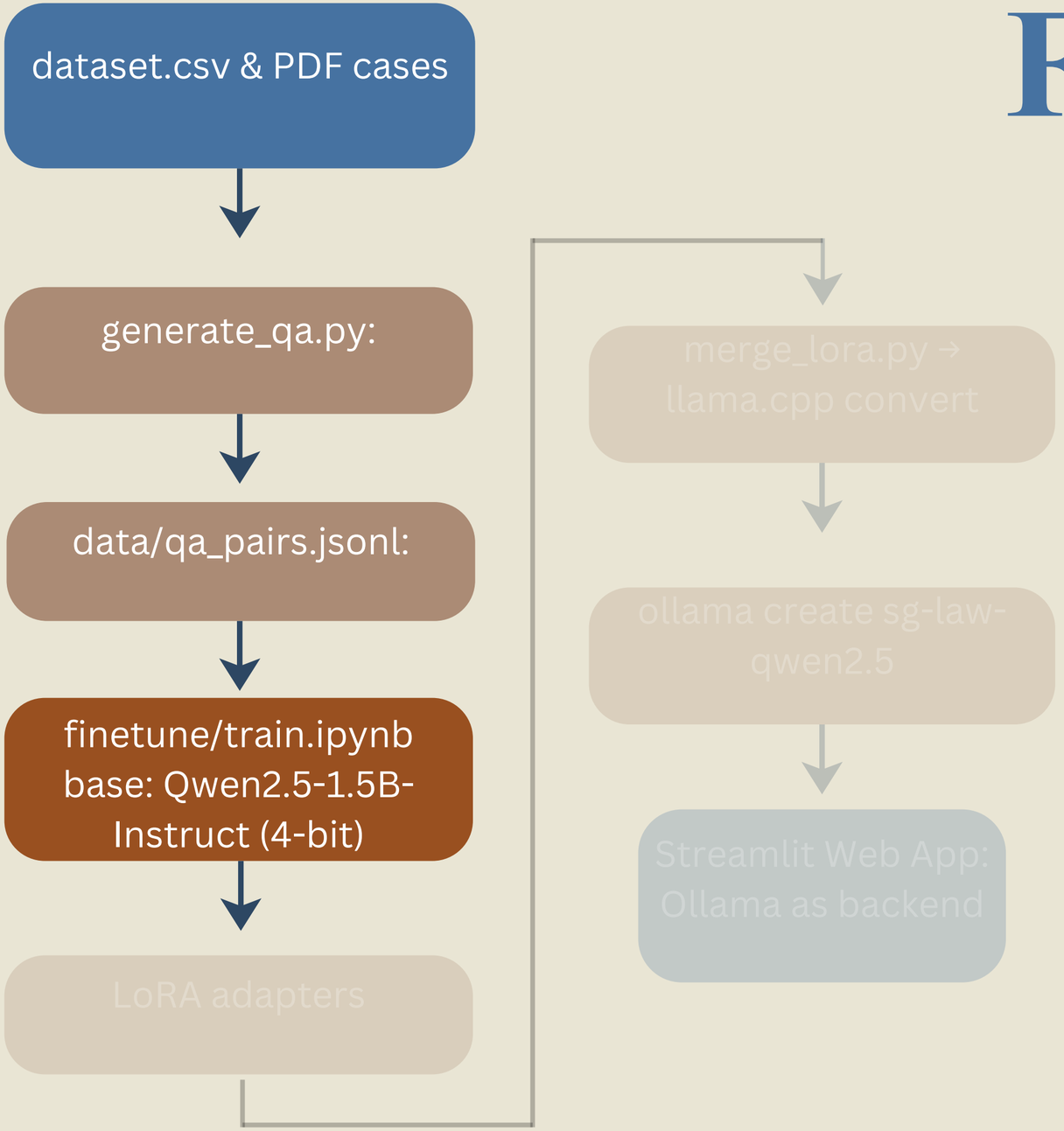
Streamlit Web



Full Pipeline

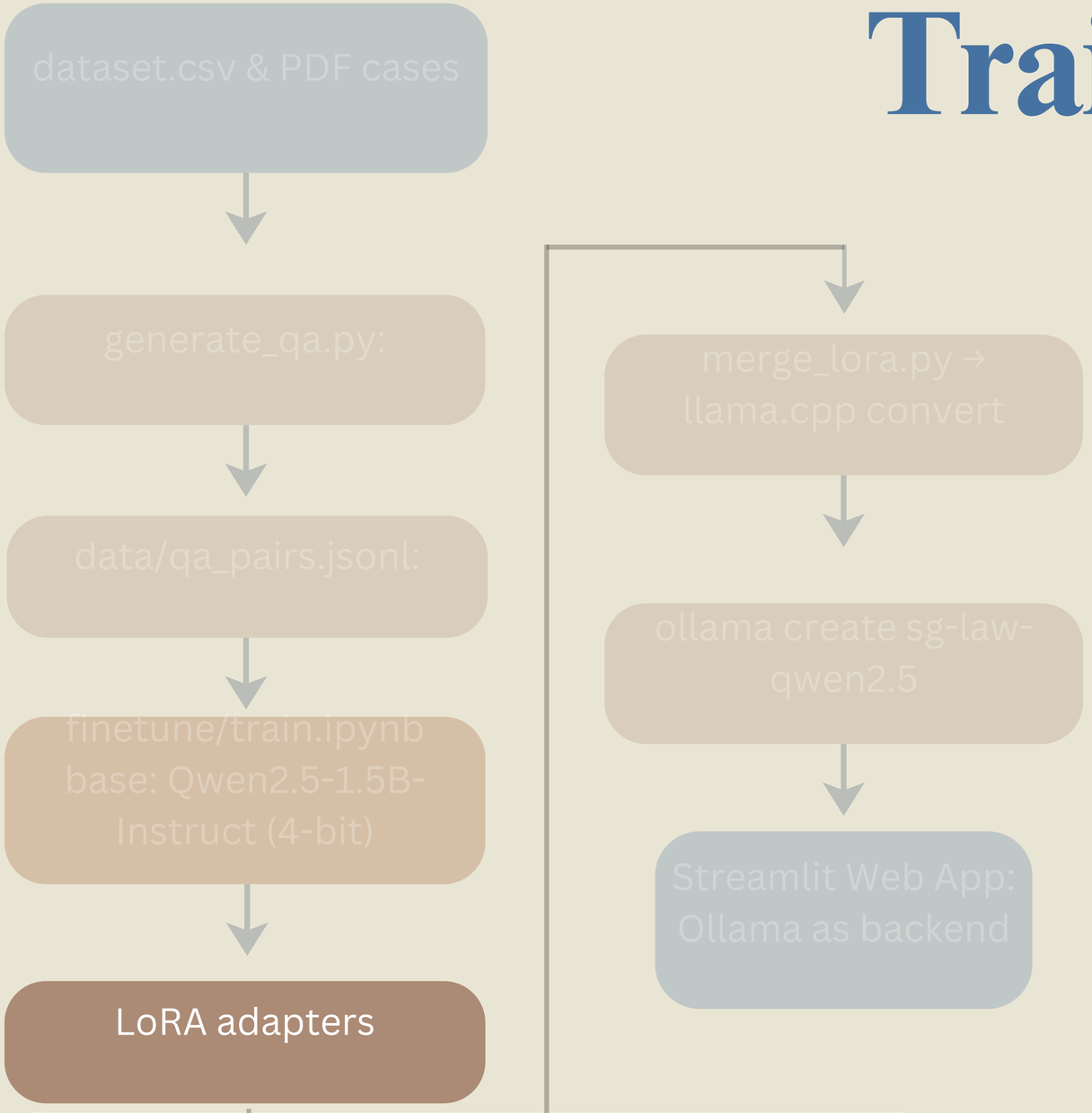


Fine-Tuning Setup



- **Base model:** Qwen2.5-1.5B-Instruct (4-bit, Unsloth)
- **Method: QLoRA fine-tuning**
- Dataset: Singapore criminal law QA pairs
  - 538 train / 60 validation samples (90/10 split)
- **Logging: Weights & Biases (W&B)**
  - Tracked loss + sample predictions

Training Configuration



EXTRA MILE · Training Techniques
Parameter-efficient fine-tuning with QLoRA (2.04% params trained)

LoRA design — efficient adaptation

- r = 16 · α = 32 · dropout = 0.05

Targets: q / k / v / o / gate / up / down_proj
Trainable params: 18.4M / 907M (2.04%)

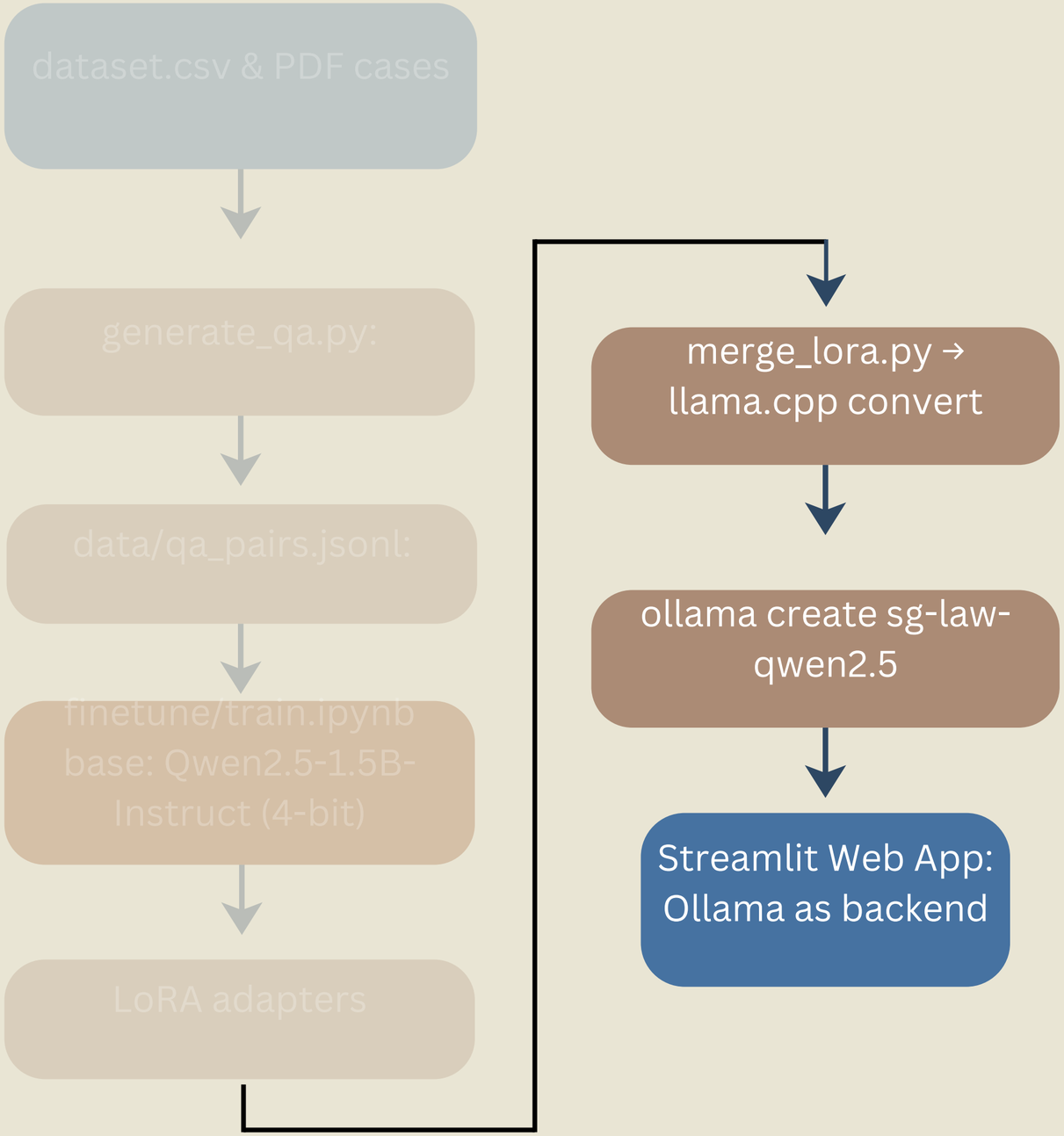
Optimisation — stable learning

- Epochs = 3 · Batch 4 × grad-accum 4 → effective 16
  LR = 2e-4 · cosine schedule · 5% warmup
  Optimizer: AdamW 8-bit · max_seq_len = 2048

Practical GPU — Colab T4 (16 GB) constraints

- 4-bit quantised base (QLoRA) + Unsloth kernels
  Training time: ~60–90 min · loss logged to W&B

Deployment



- Two deployment variants:
  - 1.5B model (~940MB, fast CPU inference)
  - 3B model (1.8GB Q4_K_M, better quality)
- Integration:
  - Auto-load in Streamlit UI model selector
- Performance:
  - Produces structured legal reasoning answers
  - Shows case-law grounding + statutory interpretation

Layer 1 — Retrieval Evaluation

20 test queries · top-5 ChromaDB retrieval · hit rate + avg cosine similarity · no API calls

0.950

hit rate

19 / 20 queries, top-5

0.664

avg cosine

mean top-5 similarity

+26.7%

vs 500-case baseline

targeted expansion → 876

Insight

property_financial
0.500 → 1.000 after targeted scrape

regulatory
stays at 0.667 — bulk of cases live in SGDC (not scraped from SUPCT)

Hit rate by domain (post-expansion)

Avg cosine similarity by domain

Layer 2 — Routing Evaluation

Manager run on 20 test queries vs hand-labelled ground truth · set-based precision / recall / F1

0.975

recall (macro)

correct domain rarely missed

0.520

precision (macro)

over-routing penalty

0.669

F1 (macro)

harmonic mean

Results by pipeline generator

Model	Prec.	Recall	F1
`sg-law-3b-q4 (fine-tuned)`	0.520	0.975	0.669
`qwen2.5:7b (base)`	0.520	0.975	0.669

Why both models score identically

The Manager applies a keyword-based domain fallback that always appends Sentencing and Criminal Procedure to any criminal query.

→ recall is near-perfect (0.975) — correct domains rarely missed.

→ precision is capped (0.520) — extra domains always appended.

Routing is dominated by the heuristic, not the LLM — so fine-tuning cannot move these scores. The real model gap appears in Layer 3 (advisory quality).

Layer 3 — Advisory Evaluation (LLM-as-judge)

Full pipeline on 10 queries · Claude used ONLY as the judge (comparison baseline — not our deployed solution) · sg-law-3b-q4 is our deployed open-source model

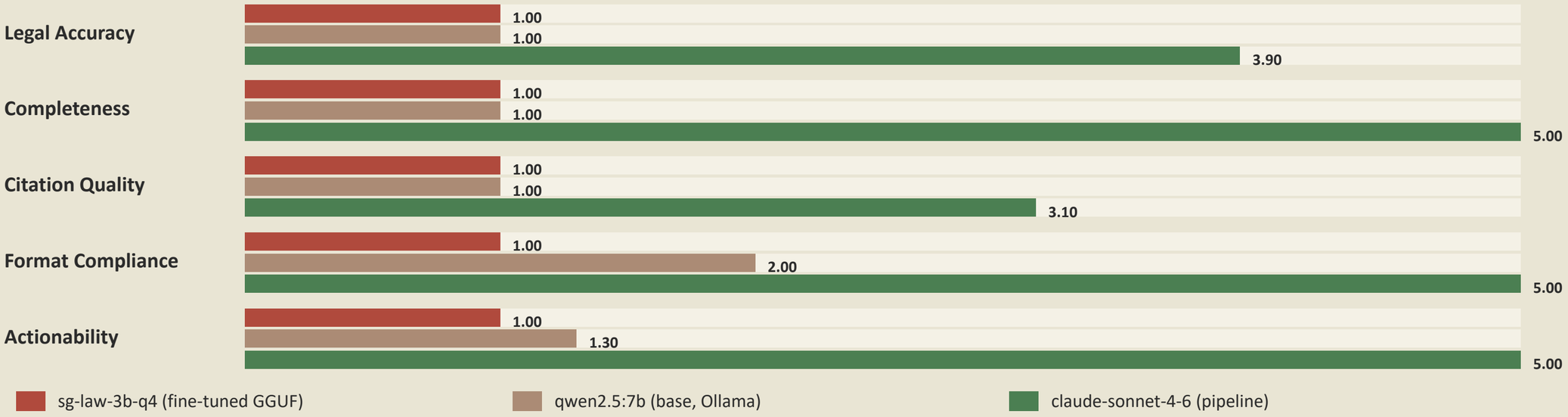
4.40	1.26	1.00
Claude (comparison)	**qwen2.5:7b base**	**sg-law-3b-q4 ← OUR SOLUTION**
claude-sonnet-4-6	Ollama local	Ollama GGUF (fine-tuned)

Adversarial rubric

Judge is prompted to deduct for wrong section numbers, hallucinated citations, missing sections, vague next steps.

Counters Claude-judging-Claude self-eval bias.

Score by dimension (1–5 scale, higher = better)



Technique Comparison — two open-source techniques, one commercial baseline

Project requires open-source / local LLM. We implemented and compared two open-source techniques; Claude is shown only as a commercial ceiling.

Technique	Model	Retrieval Hit Rate	Routing Macro-F1	Advisory (Claude-judged, /5)
OUR DEPLOYED SOLUTION **Agentic RAG + Fine-Tuned LLM**	`sg-law-3b-q4 (Ollama GGUF Q4_K_M)`	**0.950**	**0.669**	**1.00 / 5**
Open-source baseline **Agentic RAG + Base LLM**	`qwen2.5:7b (Ollama local)`	**0.950**	**0.669**	**1.26 / 5**
Commercial ceiling — NOT deployed **Agentic RAG + Claude**	`claude-sonnet-4-6`	**0.950**	**—**	**4.40 / 5**

Key takeaway

- Retrieval + routing are pipeline-level (identical across all techniques) — the honest open-source comparison is on advisory quality.
- Fine-tuning a 3B Q4-quantised model produces a smaller, locally-hostable system but hits a quality ceiling; Claude is shown only to quantify that gap.
- **Our deployed system is fully local and open-source: Qwen2.5 + QLoRA → GGUF → Ollama. No commercial API in the runtime path.**

Limitations

Source coverage

SUPCT only — SGDC (District Court) not scraped; why regulatory hit-rate stays 0.667

3B quantised

Q4_K_M cannot simultaneously hold 6-section format, correct citations, and multi-expert synthesis

Decision-support, not advisor

Output is a first-pass assessment; no accuracy guarantees, no substitute for counsel

Future Improvements

Top priority per horizon — each item directly attacks a specific limitation

Near term

Scrape SGDC

Adds District Court cases (regulatory + road-traffic) — directly fixes the 0.667 regulatory hit-rate gap

Mid term

Learned router

Replace keyword fallback with a small classifier — unlocks real routing precision instead of the always-appended Sentencing + Criminal Procedure

Long term

Scale base model

Fine-tune 7B / 14B for full advisory synthesis — 3B Q4_K_M is a hard quality ceiling

Q&A